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SIXTH SEMESTER

Roll No. 2120/CO/035

B.Tech. [CSE]

MID SEMESTER EXAMINATION *March-2023*

CO304 ARTIFICIAL INTELLIGENCE

Time: 1:30 Hours *Max. Marks: 25*

Note:	Answer *ALL* questions. Assume suitable missing data, if any. CO# is course outcome(s) related to the question. L# is the cognitive level required to solve the question.

1[a] Consider 3-puzzle problem, it consists of a 2×2 board with three numbered tiles and a blank space. A tile adjacent to the blank space can slide into the space. The object is to reach a specified goal state(s). The initial state S goal states G_1 and G_2 are shown in Fig. 1. Give the problem formulation for the above problem. Illustrate the state space with the help of a diagram, along with applicable actions. Also, give a possible sequence of actions to achieve goals G_1 and G_2, if there are any.

[1+2+2] [CO1, CO2] [L3]

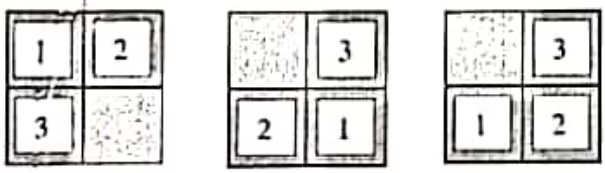

Fig.1: (a) S, (b) G_1, (c) G_2

[b] Consider a vacuum-clear World shown in Fig. 2. This World has just three locations: rooms A, B and C. The vacuum agent perceives which room it is in and whether there is dirt. It can move left, move right, suck up the dirt, or do nothing. One straightforward agent function is the following: if the current room is dirty, then suck; otherwise, move to the other room. Find the number of states in the state space and write a pseudo code for the reflex agent program.

[1+2] [CO2] [L3]

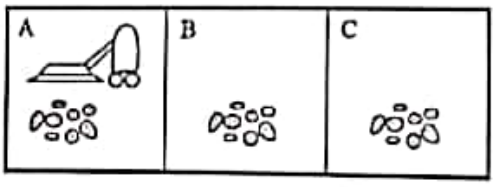

Fig. 2

2[a] Consider the state space shown in Fig. 3. Numbers on the edges show the step path cost between the nodes. Use the breadth-first search algorithm to find the path from start state S to a goal state G. What should be the step path cost(instead of given values) if the optimal path is desired using the breath-first search? Show each step clearly. Answer with limited words; you may use diagrams for an explanation. Do not use large text paragraphs.

[2+1] [CO1, CO2] [L3]

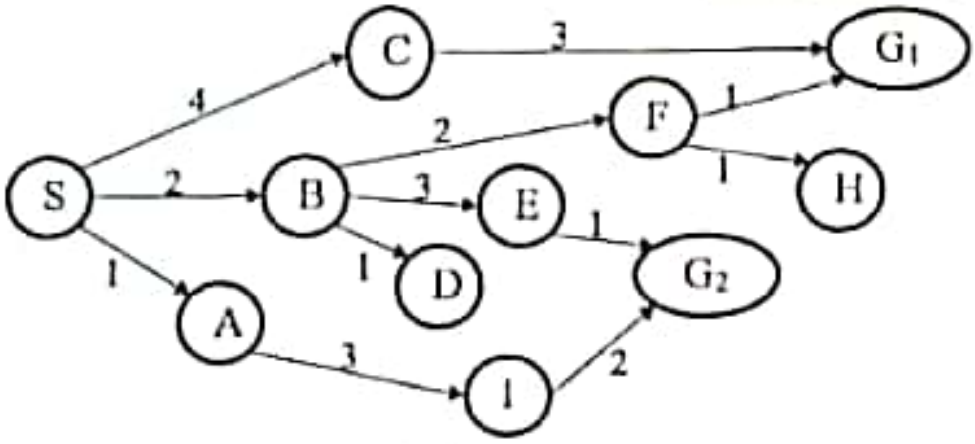

Fig. 3

[b] Consider the state space shown in Fig. 3. Find the optimal cost path from the start state to a goal state using the uniform cost search algorithm. Compare the path cost obtain in part 2[a]. Also, explain why the difference in two path costs is given by breath-first and uniform cost search algorithms (use a maximum of three lines for explanation). [Note: if equal cost nodes are generated at any stage, the expansion order will be first-in-first-out (FIFO)]. **[2+1] [CO2] [L4]**

3[a] Consider two players playing a game. The game's setup is on a map and two pieces of knights (from chess): to indicate their positions on the map. Players can simultaneously move to a neighbouring city on the map at every turn. The time needed to move from city i to neighbour j is equal to the road distance $d(i, j)$ between the cities. The players take the next turn when both complete the previous turn. We want the two players to meet as quickly as possible. Let $D(i, j)$ be the straight-line distance between cities i and j. Which of the following heuristic functions is/are admissible? (i) $D(i, j)$;

(ii) $2 \cdot D(i, j)$;

(iii) $D(i, j)/2$.

Give one or two lines of explanation for each. **[2] [CO3] [L2]**

[b] Consider the state-space shown in Fig. 4. The values of the heuristic function at each node are given in Table I. Find the path from start node S to the gaol node G using A* search algorithm (i) tree search version, (ii) Graph search version. Will it give the optimal path and the same path in both cases? If not, which one is an optimal path? Why does any one (or both) search versions give a sub-optimal path?

[2+2+2] [CO3] [L3, L4]

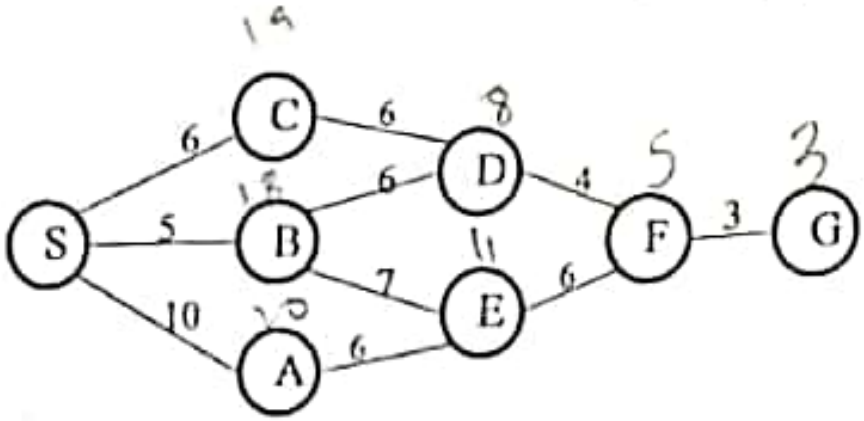

Fig. 4

Table I

Node (n)	S	A	B	C	D	E	F	G
Heuristic $h(n)$	17	10	13	4	2	4	1	0

4 Consider the blocks World problem shown in Fig. 5. It consists of cubes with alphabets. Some of the blocks are placed on a table. There is a robot arm to pick up or put down the blocks. The robot arm can move only one block at a time from the top of the stack or table. The aim is to achieve the goal state from the start state. Illustrate where the simple hill-climbing algorithm can be used to achieve the goal state. If not, why? Show the steps of the simple hill-climbing. Use heuristic function h.

h: for each block at the desired height (with respect to goal state), it adds the number of blocks underneath it (for example: in goal state, for block C it adds 2, and for block E, it adds 0; thus, h at goal state = 4+3+2+1+0 = 10). For each block at wrong height (with respect to goal state), it adds minus times the number of block positions displaced above or below (for example: in start stat, for block C it adds -1, and for A, it adds -4)

Fig. 5

Investigate whether the steepest ascent version of hill climbing will achieve the goal. **[2+1] [CO3] [L3, L4]**

—Best of Luck—